Location Over Space: Madrid's Rental Market Analysis

Clustering Segmentation and Regression Pricing Model

Group 8: Alberto Cabezudo Rodríguez, Madelyn Ehni, Gilles Hamers, Nicolás A. Higuera Wilches, Salah Mneimne

Machine Learning I February 28, 2026

Executive Summary

This report presents a two-phase machine learning analysis of rental listings from *Idealista.com* across Madrid. In **Phase 1**, a K-Means[1] clustering model segments properties into structurally distinct sub-markets based on the number of bedrooms and floor level, achieving four well-defined property 'profiles' (Silhouette[2] = 0.455). In **Phase 2**, a linear regression model estimates rental prices within the largest segment, *Compact Traditional*, which represents ≈43% of listings. This model achieved an R^2[3] of 0.644 and MAPE[4] below 20% across training, testing, and reserved datasets. Together, they answer two key questions for every listing: which market segment it belongs to, and what it should rent for.

Madrid's rental market is made up of four distinct sub-markets. The clustering analysis revealed that property type and floor level define the market structure more clearly than any combination of amenities. Our resulting segments span a € 1,300 median monthly rent range with *Compact Traditional* properties anchoring the majority of the market, followed by *Small High-Rise Modern* properties with a 27% median rent premium over it, even with an identical number of bedrooms. This gap is driven by floor-level and building modernity. At the top of the market, *Large Low-Rise Family* and *Large High-Rise Executive* residences share a median rent price of € 2,300 and € 2,400 but serve opposite tenant needs: garden access and suburban versus height, penthouse access and luxury districts. Pricing these two segments against a shared benchmark systematically misprices both.

Within the *Compact Traditional* segment, the highest volume market, location is the dominant influence over rent, not size. The regression model shows that the marginal value of a square meter varies distinctly by district: in Salamanca, each additional m² is worth € 18.2 in rent, 2.5 times the € 7.4 baseline rate observed in peripheral districts. Outer districts such as Puente de Vallecas carry a structural discount regardless of property attributes. An identical 70 m^2 property can cost € 300–€ 400 more per month just based on neighborhood. Model performance is stable across training, testing, and hold-out sets, confirming that its pricing predictions will hold on new listings, not just the data it was trained on.

1. Data & Methodological Framework

1.1 Data Audit & Cleaning

The raw dataset contains 15 variables and 2,089 observations. Five variables carried missing values, three reported extreme values (outliers), and one registered a potential data-entry error.

Listings with missing values for Bedrooms and Floor were imputed using a 'contextual' median, based on their respective District and Area to preserve neighborhood-specific building patterns. Meanwhile, missing values for Elevator and Outer were handled using mode imputation.

Extreme values or outliers were defined using the IQR[5] method. Listings with extreme rent prices and floor areas were capped with upper limits of € 4,825 and 270 m^2 respectively, and bedrooms count was capped at four. Finally, a floor value of 43,039 (an obvious data-entry error) was flagged and treated as a missing value.

A full summary of quality issues and corrective actions is provided in Table A1.

1.2 Clustering Variable Selection

Seven candidate variables were identified (see Annex 1). Here is what we discovered:

Rent and District were excluded from the clustering: Rent serves as the objective for the Phase 2 regression analysis. Including either option would distort segmentation in favor of pricing results, as the district anticipates regression.

Multicollinearity screening revealed a strong correlation between Bedrooms and square meters (r=0.755; VIF[6] $\approx$ 9) as shown in Table A2. Bedrooms + Floor (VIF 2.29) and Sq.Mt. + Floor (VIF 2.09) emerged as the two most effective combinations. Given that bedrooms serve as the main filter for tenant searches and exhibited the highest silhouette score (0.4552 compared to 0.4053), we opted for Bedrooms + Floor.

[1] **K-Means:** Unsupervised algorithm that partitions observations into K groups by minimizing squared distances to cluster centroids.

[2] **Silhouette score:** Ranges from -1 to $+1$; measures how similar each observation is to its own cluster vs. the nearest alternative. Values above 0.4 indicate meaningful structure.

[3] **R² (Coefficient of Determination):** Proportion of rent variation explained by the model; 0.644 means 64.4% of price differences are accounted for by the predictors.

[4] **MAPE (Mean Absolute Percentage Error):** Average prediction error as a percentage of actual value. A MAPE of 20% means predictions are, on average, within 20% of the true rent.

[5] **IQR (Interquartile Range):** Difference between the 75th and 25th percentiles. Values beyond $1.5\times$ IQR above Q3 are flagged as outliers.

[6] **VIF (Variance Inflation Factor):** Quantifies how much a predictor's variance is inflated by collinearity with other predictors. VIF < 5 is acceptable; VIF ≥ 10 requires removal.

Finally, binary amenities (Elevator, Outer, Penthouse, Cottage) showed 3–5× standard-deviation separation after standardization. Including them produced *dominated clusters*. Including individual amenities rather than focusing on market structure will lead to the formation of risk clusters, so they were kept as profiling variables.

2. Technical Approach

2.1 Algorithm Choice & K Selection

K-Means was chosen for three reasons: **(a)** The clusters are interpretable. A 3.5-bedroom centroid on floor 6.7 communicates a real property type. **(b)** The model is fully reproducible on updated listings at any time and get consistent, comparable segments. **(c)** Each property's cluster label is used to filter the dataset, so the regression model trains only on properties from that segment.

All variables were standardized (z-score) before fitting. The number of potential groups or clusters (K) was evaluated from 2 to 7 using four metrics: Inertia, Silhouette, Calinski-Harabasz, Davies-Bouldin. Full results are shown in Table A3.

$K=3$ delivers the highest Silhouette (0.460) but merges the Large High-Rise Executive and Large Low-Rise Family segments. However, $K = 4$ (Silhouette 0.455) was selected because it preserves the structural distinction between large high-rise and large low-rise properties. The 0.0047 Silhouette trade-off is commercially justified.

2.2 Scenario Testing

As illustrated in Table A4, we tested twelve different variable combinations at K=4 to verify whether binary amenities should be included in the clustering. The silhouette score, variance inflation factor (VIF), cluster balance (minimum/maximum percentage), and a domination check were used to evaluate each scenario.

While several scenarios (3–12) show relatively high silhouette values, this enhancement is misleading. In such cases, the algorithm categorizes listings mainly by whether that specific feature is present or absent, rather than following a logical market structure, as the distance calculation is largely influenced by a binary amenity.

This is particularly clear in Scenario 7, where 68.4% of the observations converge into one cluster. Only Scenario 1 yields segments that are both economically interpretable and balanced, all while fulfilling the necessary validation criteria.

3. Segmentation Results

3.1 Model Validation

The final model was validated across four independent validation criteria (n (%), Silhouette, Negative Silhouette, Rent CV (σ/μ)) before any business interpretation was applied and can be found in Table A5.

Table 1: Cluster validation summary ($K = 4$, Silhouette = 0.455).

Cluster	n (%)	Sil.	Rent CV	Bal.
C0 Large Hi-Rise Exec.	229 (11.0%)	0.370	0.478	PASS
C1 Compact Traditional	903 (43.2%)	0.476	0.532	PASS
C2 Large Lo-Rise Family	624 (29.9%)	0.489	0.537	PASS
C3 Small Hi-Rise Modern	333 (15.9%)	0.394	0.487	PASS

All four clusters meet the validation checks. Rent dispersion remains stable (CV between 0.478 and 0.537), only one listing shows a negative silhouette, and cluster sizes are well balanced. In simpler terms, almost every property sits closest to its assigned centroid.

C1 (*Compact Traditional*) represents 43.2% of all listings, has zero negative silhouettes, and a silhouette score of 0.476. Its rent dispersion (σ €677; CV 0.532) is substantial, making it the most commercially relevant segment for Phase 2 regression modeling.

What is also noticeable is that the median rents confirm a clear price tier separation within a range of € 1,300: The lowest is **C1** at € 1,100, followed by **C3** at € 1,400 (+27%), then **C2** at € 2,300 (+64%) and **C0** with a median rent of € 2,400.

3.2 Cluster Profiles

The 2×2 matrix below captures the structural segmentation logic of the 4 clusters within two dimensions: Unit size (compact ≤2 beds vs. large ≥3 beds) and floor level (low-rise ≤4 vs. high-rise ≥5). Detailed Cluster Profiles, Pricing Mechanisms and Tenant Personas are reported in Table A6 to support the interpretation.

Table 2: Segmentation matrix: 4 clusters in 2 dimensions.

	Low-Rise ($\bar{F}\approx 2$)	**High-Rise ($\bar{F}\approx 6.7$)**
Large (3–4 beds)	**C2** *Lg. Low-Rise Family* $n=624$ (29.9%) Med. rent € 2,300 € 13.7/m^2 | Cottage 13%	**C0** *Lg. Hi-Rise Exec.* $n=229$ (11.0%) Med. rent € 2,400 € 15.0/m^2 | Penthouse 20%
Compact (1–2 beds)	**C1** *Compact Traditional* $n=903$ (43.2%) Med. rent € 1,100 € 16.2/m^2 | Elev. 82%	**C3** *Small Hi-Rise Modern* $n=333$ (15.9%) Med. rent € 1,400 € 17.7/m^2 | Penthouse 23%

Selected segment profile: *Compact Traditional* (**C1**) listings cover smaller apartments on lower floors in older city buildings, many without full lift access. This is the most common segment in the market. Rents are relatively affordable in total, but expensive per square meter because of the central location. What matters most here is the street and neighborhood, not extra space or luxury features. This cluster was selected for our regression model for three reasons: It represents 43.2% of all listings, it carries the highest within-segment rent dispersion (CV = 0.532, σ = €677), and its €16.2/m^2

price intensity points to a location-driven pricing pattern that a district-aware model can directly exploit.

3.3 Business Interpretation

(a) *Compact Traditional* C1 is the agency's most important segment. At 43.2% of all listings, it's the most common category on any platform. An apartment in Centro costs between €300 and €400 more per month than an identical apartment in Vallecas, despite a median rent of €1,100 and a standard deviation of €677. However, without a model that accounts for district variations, apartment prices are similar. Compact Traditional apartments' price intensity of €16.2/m^2 outperforms the two larger unit segments, C2 at €13.7/m^2 and C0 at €15.0/m^2, as tenants prioritize central location over space.

(b) *Large High-Rise Executive* (C0) and *Low-Rise Family* (C2) share a price point but are opposite products. In both clusters, 160 m^2 three-bedroom homes rent for €2,300–€2,400 per month. **C0** offers remarkable height, stunning views, and a luxurious interior (Penthouse 19.7%, Elevator 99.6%, Floor mean 6.7, Salamanca/Chamartín). **C2** has garden access and suburban dimensions (Moncloa/Hortaleza, Cottage 13.1%, Floor mean 2.3). Using a large-unit benchmark for both will misprice one. All of this shows that $K = 4$ emphasizes this difference, while $K = 3$ minimizes it.

(c) *Small High-Rise Modern* (C3) vs. *Compact Traditional* (C1) reveals the floor premium. Both clusters have a median of 2 bedrooms, yet **C3** rents for €1,400/month vs. **C1**'s €1,100/month, which is a 27.3% premium attributable entirely to floor level (6.7 vs 2.1) and modernity. **C3**'s €17.7/m^2 is the highest price intensity in the dataset. An agent benchmarking a **C3** listing against a comparable **C1** leaves roughly €300/month on the table per listing. The correct comparison is floor-matched, not bedroom-matched.

4. Linear Regression Model

4.1 Scope & Setup

For the linear regression model, we focus on **C1** (*Compact Traditional*), which includes 903 properties (≈43% of the dataset), making it the largest and most statistically robust segment. This cluster is characterized by compact apartments, typically with one or two bedrooms and located on relatively low-to-mid floors. Compared to other clusters, it combines a high number of observations with moderate dispersion in rental prices, providing a stable basis for modeling and reducing structural heterogeneity (see Table 1).

Non-predictive columns are then removed, including Id, Address, Number, Area, and property-type variables such as Cottage and Semidetached, which either lack explanatory value or are not representative within this cluster. The remaining variables are classified as continuous (e.g., Sq.Mt, Bedrooms, Floor) or binary (e.g., Outer, Elevator, Penthouse, Duplex). Histograms show moderately right-skewed distributions in size and rent, typical of housing markets, but without extreme tails after outlier treatment (see Figure A7). Binary variables display sufficient variability to contribute meaningfully to regression estimation (see Figure A8).

Finally, special treatment is applied to District. To avoid unstable coefficients from sparsely represented locations, districts with fewer than 25 listings are grouped into an "Other" category (see Table A7). One-hot encoding with a reference district is then implemented, ensuring interpretability while preventing multicollinearity. At this point, the dataset is structurally consistent and ready for OLS[7] modeling.

4.2 Model Building & Selection

To ensure robust evaluation, the dataset is divided into three subsets: 10% is set aside as a reserved (hold-out) set, while the remaining 90% is split equally into training and testing sets. The training set is used for estimation and specification adjustments; the testing set evaluates intermediate refinements, and the reserved set provides a final unbiased assessment. This structure allows us to prevent overfitting and improves generalization reliability.

The baseline OLS model with all predictors achieved $R^2 = 0.627$ (Adj. $R^2 \approx 0.607$), explaining a substantial share of rent variation within **C1** (see Table A8). Bedrooms and Floor, our clustering variables, were dropped with virtually no R^2 loss (≈0.626), confirming their limited variability within the cluster (see Table A9).

Multicollinearity is assessed through the pairwise correlation matrix and Variance Inflation Factor (VIF), with a VIF threshold of 10, indicating acceptable levels as shown in Table A10.

Feature selection via RFECV[8] reduced the model to 17 predictors; backward elimination further removed non-significant district dummies such as Hortaleza and Ciudad Lineal.

To enhance interpretability, interaction terms between Sq.Mt and District are introduced. These allow the marginal effect of size to vary by location, capturing spatial heterogeneity. The final model improves to an R^2 of 64.4%, with adjusted R^2 closely aligned, indicating genuine explanatory gains rather than overfitting. Districts such as Puente de Vallecas and the grouped Other category show negative base effects, while several central districts display positive size interaction effects, meaning square meters are valued differently across locations (Table A11).

4.3 Final Model and Performance Metrics

The final model retains 10 features: Sq.Mt, Outer, two district dummies (Other, Puente Vallecas), and six Sq.Mt × District interactions. The implications of the model's output are discussed in the next section.

[7] **OLS (Ordinary Least Squares):** Linear regression method that estimates coefficients by minimizing the sum of squared residuals between observed and predicted values.

[8] **RFECV (Recursive Feature Elimination with Cross-Validation):** Iteratively removes the least important predictor and uses cross-validation to identify the optimal feature subset.

Table 3: Final OLS model — coefficient estimates (Cluster 1, $n_{\text{train}} = 406$).

Predictor	$\hat{\beta}$	Std. Err.	t	Sig.
Base predictors				
Intercept	375.39	63.26	5.94	***
Sq.Mt (m^2)	7.38	0.95	7.78	***
Outer (exterior unit)	107.27	51.84	2.07	*
District: Other	−214.98	79.51	−2.70	**
District: Puente Vallecas	−199.91	99.25	−2.01	*
Sq.Mt × District interactions (marginal €/m^2 over baseline)				
× Centro	6.03	0.73	8.31	***
× Chamartín	3.79	0.99	3.85	***
× Chamberí	5.78	0.96	6.02	***
× Moncloa	3.22	1.41	2.28	*
× Retiro	4.26	1.20	3.57	***
× Salamanca	10.85	0.83	13.12	***

*** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$
$R^2 = 0.644$, Adj. $R^2 = 0.635$, $F(10, 395) = 71.53$, $p < 0.001$

Table 4: Regression performance across data splits.

Metric	Train	Test	Reserved
R^2	0.644	0.644	0.630
MAE (€)	260	267	242
RMSE (€)	391	426	364
MAPE (%)	19.7	20.2	19.2

4.4 Model Interpretation

The model explains nearly two-thirds of rent variation within *Compact Traditional*. An R^2 of 0.644 means that district, size, and their interaction together account for 64.4% of the price differences observed across these properties. The remaining 35.6% reflects factors the model cannot observe from a listing alone such as interior quality, renovation standard, specific floor orientation, and landlord negotiation.

The model's coefficients reveal the pricing mechanics of Madrid's *Compact Traditional* market:

- **Base rate:** €375/mo for a hypothetical 0 m^2 non-exterior unit in the reference district—the intercept anchors the model.
- **Size premium:** Each additional m^2 raises average rent prices by €7.4 in the reference district, holding other attributes constant.
- **Location multiplier (Sq.Mt × District):** Not only size matters, location is key. The increase in monthly rent for an additional m^{2}changes significantly across districts. The total value of this increase equals the baseline (€7.38) plus the district's interaction. Salamanca leads with a marginal value of (€10.8/m^2) on top of the base, followed by Centro (+€6.0) and Chamberí (+€5.8). A 70 m^2 flat in Salamanca is predicted at ∼€1,650, vs. ∼€890 in the reference district, an 85% location premium.
- **Peripheral discount:** In contrast, Puente Vallecas (−€200) and "Other" grouped districts (−€215) carry negative base effects, reflecting lower demand.
- **Exterior premium:** Outer-facing units add €107 ($p = 0.039$), a modest but significant amenity premium.

The key insight: Within *Compact Traditional*, **location dominates size**. The same 70 m^2 unit can range from €890 to €1,650 depending on district, a spread larger than what any physical feature can produce. This confirms the clustering finding that **C1 tenants "pay for address, not space".**

The model will perform as reliably on new listings as it does on the data it was trained on (see Figure A9). R^2 holds at 0.64 across training and test sets and falls only to 0.63 on the reserved hold-out. Predictions will not deteriorate as new listings enter the portfolio.

5. Conclusion & Recommendations

The Madrid rental market is segmented, not continuous. Treating all listings as a single market creates systematic pricing errors. The segmentation identifies four discrete property types with distinct price dynamics. The *Compact Traditional* segment accounts for 43% of all listings and should anchor the agency's volume strategy. It is large enough to justify a dedicated pricing process and small enough to model with precision.

Where a property is located matters more than how big it is. Within the *Compact Traditional* segment, district drives rent more than floor area. A square meter in Salamanca is €10.85 above the base rate, nearly 2.5× the premium of a square meter in the average district. Pricing primarily by size leaves a margin on the table in premium locations and over-pricing in peripheral ones.

Segment-first pricing prevents the errors a blended model cannot see. Two segments (**C0** and **C2**) occupy the same price range with opposite profiles. A single-market model would average them out and produce wrong estimates for both. Segmenting first ensures the right pricing logic is applied to the right property type.

5.1 Recommendations

- **Classify before you price.** Use the clustering model to assign each new listing to a segment, then apply the corresponding regression model. A one-step pricing tool built on a blended model will underperform for most property types.
- **Use the model as a mispricing detector, not just a price setter.** Flag any listed property whose asking rent falls more than 1σ below the model's prediction; these are the clearest candidates for upward renegotiation.
- **Treat the model's output as a floor, not a ceiling, at the top end.** Above €2,000/month the model under-predicts, because prestige-street effects and interior specification are not captured. For premium listings, use the predicted rent as the minimum justifiable price.
- **Extend the pipeline to the remaining three segments.** Dedicated regression models for C0, C2, and C3 would cover the remaining 57% of the portfolio with the same precision achieved here for C1.

Technical Annex

Annex 0 — Data Cleaning Summary

Table A1: Data quality issues and corrective actions.

Variable	Issue	Scale	Resolution
Floor	Entry error: value = 43,039	1 record	Set to missing; contextual median imputation
Rent	Right-skewed; max € 16,000	114 rows (5.5%)	IQR cap at € 4,825
Sq.Mt	Right-skewed; max 1,250 m^2	188 rows (9.0%)	IQR cap at 270 m^2
Bedrooms	Outliers up to 8; 89 missing	162 capped; 89 imputed	Cap at 4; contextual median (District+Area)
Floor	142 missing post error fix	142 rows (6.8%)	Contextual median: Dist.+Area → Dist. → Global
Elevator / Outer	133 / 162 missing	6.4% / 7.8%	Mode imputation (both mode = 1)

Annex 1 — Data Audit & Variable Selection

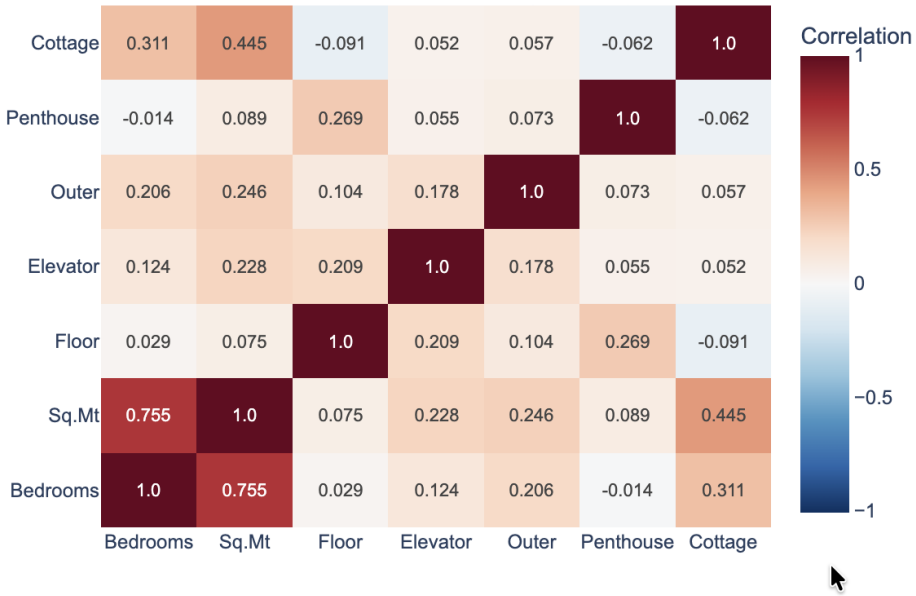


Figure A1: Pairwise correlation matrix for candidate clustering variables. Strong correlation between Bedrooms and Sq.Mt ($r = 0.755$) confirms their redundancy; Bedrooms and Floor ($r = 0.044$) and Sq.Mt and Floor ($r = 0.075$) are near-independent, validating the final two-variable selection.

Table A2: VIF screening of candidate variable combinations.

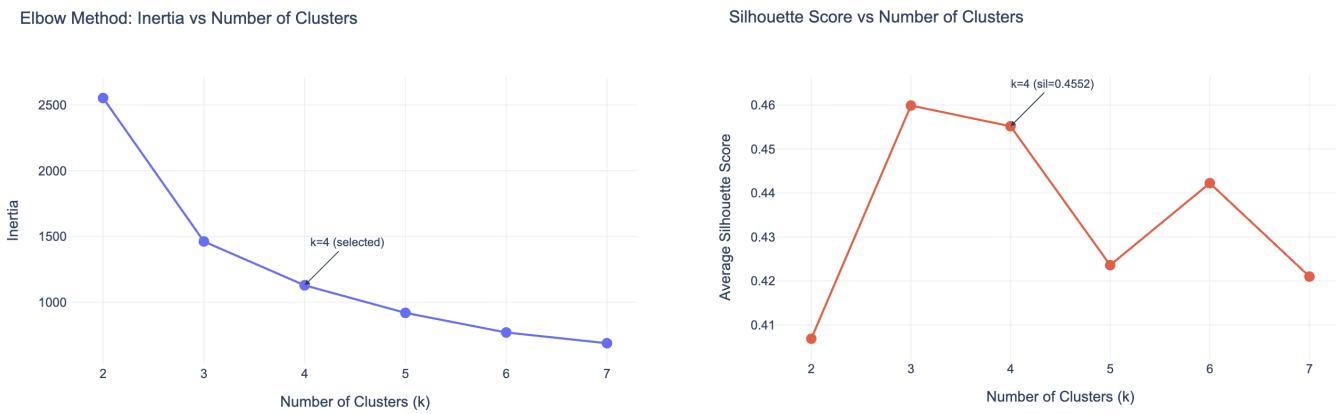
Combination	Variable	VIF	Status
Bedrooms + Sq.Mt + Floor		*max 9.88*	
	Bedrooms	9.88	↑ high
	Sq.Mt	9.01	↑ high
	Floor	2.30	ok
Bedrooms + Floor		*max 2.29*	
	Bedrooms	2.29	ok
	Floor	2.29	ok
Sq.Mt + Floor		*max 2.09*	
	Sq.Mt	2.09	ok
	Floor	2.09	ok

Conclusion: Bedrooms and Sq.Mt cannot coexist (VIF > 5). Two viable base combinations advance to Stage 4: **Bedrooms + Floor** and **Sq.Mt + Floor**. Threshold: VIF < 5 acceptable; VIF < 10 tolerable.

Annex 2 — Algorithm Choice, K Selection & Scenario Testing

Table A3: Cluster validation metrics ($K = 2$ to 7).

K	Inertia	Silhouette	Calinski-H.	Davies-B.	Decision
2	2,553	0.407	1,328	1.077	Too coarse
3	1,461	**0.460**	**1,939**	0.790	Statistically strongest
4	**1,127**	**0.455**	1,880	0.837	**SELECTED**
5	918	0.424	1,851	0.875	Diminishing returns
6–7	$<$769	$<$0.443	$<$1,848	$>$0.848	Over-fragmented



(a) Elbow plot: Inertia vs. K.

(b) Silhouette and additional metrics vs. K.

Figure A2: K-selection diagnostic plots.

Table A4: Twelve-scenario variable-combination testing at $K = 4$. *Green* = selected | *Pink* = dominated cluster detected.

#	Variable Combination	Vars	Silhouette	VIF_{max}	Min Cluster	Max Cluster	Dominated
Bedrooms-based combinations							
1	Bedrooms + Floor	2	**0.4552**	2.29	11.0%	43.2%	No
2	Bedrooms + Floor + Elevator	3	0.4751	5.63	11.2%	37.4%	Yes
3	Bedrooms + Floor + Outer	3	0.4718	5.46	12.3%	36.7%	Yes
4	Bedrooms + Floor + Penthouse	3	0.4825	2.52	8.1%	43.7%	Yes
5	Bedrooms + Floor + Cottage	3	0.4680	2.60	4.2%	45.4%	Yes
6	Bedrooms + Floor + ALL amenities	6	0.4812	7.25	4.2%	68.4%	Yes
Sq.Mt-based combinations							
7	Sq.Mt + Floor	2	0.4053	2.09	13.6%	34.5%	No
8	Sq.Mt + Floor + Elevator	3	0.5092	5.02	11.2%	47.4%	Yes
9	Sq.Mt + Floor + Outer	3	0.5089	4.60	12.3%	48.2%	Yes
10	Sq.Mt + Floor + Penthouse	3	0.5180	2.24	8.1%	55.7%	Yes
11	Sq.Mt + Floor + Cottage	3	0.5043	2.65	4.2%	56.2%	Yes
12	Sq.Mt + Floor + ALL amenities	6	0.5141	7.17	4.2%	77.1%	Yes

Note: Higher silhouette in scenarios 3–12 is misleading — a binary amenity dominates the distance calculation, splitting listings on presence/absence of a single feature rather than market structure. Only Scenario 1 satisfies all criteria: balanced clusters, no domination, VIF < 5, and economic interpretability.

Annex 3 — Segmentation Results

Table A5: Final model validation criteria.

Cluster	n (%)	Sil.	Neg. Sil.	Rent CV (σ/μ)	Balance
C0: Large Hi-Rise Exec.	229 (11.0%)	0.370	1	0.478 (mean € 2,590)	PASS
C1: Compact Traditional	903 (43.2%)	0.476	0	0.532 (mean € 1,273)	PASS
C2: Large Lo-Rise Family	624 (29.9%)	0.489	0	0.537 (mean € 2,448)	PASS
C3: Small Hi-Rise Modern	333 (15.9%)	0.394	0	0.487 (mean € 1,728)	PASS

Annex 4 — Cluster Interpretation

Table A6: Full cluster profiles, pricing logic, and tenant personas.

Cluster	Size	Physical Profile	Rent Profile	Amenities	Top Districts	Pricing Logic	Tenant Persona
C0 Lg. Hi-Rise Exec.	229 (11%)	3 beds, 6, 160 m^2 Floor	€2,400, €15.0/m^2	Elev. 99.6%, Pent. 19.7%	Salamanca 25%, Chamartín 17%	Height + penthouse + prestige district	Executive families, corporate relocations
C1 Compact Trad.	903 (43%)	2 beds, 2, 70 m^2 Floor	€1,100, €16.2/m^2	Elev. 81.6%, Pent. 3.0%	Centro 22%, Chamberí 10%, Tetuán 10%	Centrality premium; address over space	Students, young professionals
C2 Lg. Lo-Rise Fam.	624 (30%)	3 beds, 2, 160 m^2 Floor	€2,300, €13.7/m^2	Elev. 90.5%, Cottage 13.1%	Salamanca 16%, Moncloa 14%, Hortaleza 11%	Space + garden access + suburban scale	Established families
C3 Small Hi-Rise Mod.	333 (16%)	2 beds, 6, 80 m^2 Floor	€1,400, €17.7/m^2	Elev. 97.9%, Pent. 23.4%	Salamanca 23%, Chamberí 13%, Chamartín 12%	Floor + modernity premium; highest €/m^2	Urban professional couples

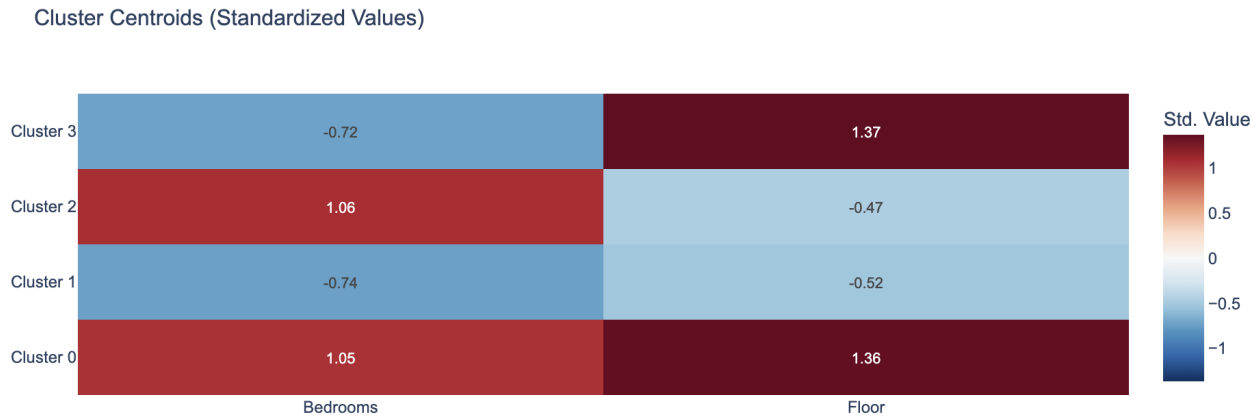


Figure A3: Cluster centroids — standardised values. The heatmap reveals a clean 2×2 structure: Clusters 0 and 2 are large-bedroom segments (+1.05–+1.06 SD) diverging sharply on floor level (+1.36 vs. −0.47), while Clusters 1 and 3 are compact-bedroom segments (−0.72–−0.74 SD) showing the same high/low floor split. No centroid is near zero on both axes, confirming each cluster occupies a distinct position in the feature space.

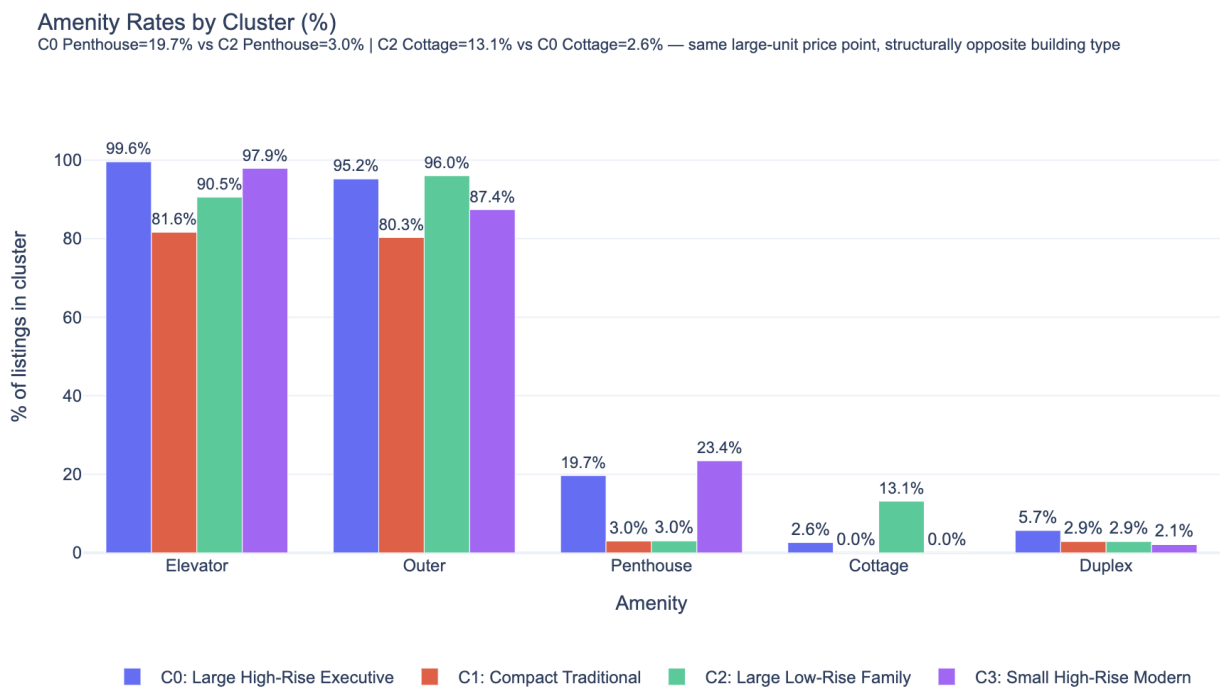


Figure A4: Amenity rates by cluster (%). Elevator and Outer are near-universal across all segments and carry limited discriminating power. The structurally opposite profiles of C0 and C2 are most visible in Penthouse (19.7% vs. 3.0%) and Cottage (2.6% vs. 13.1%) — two clusters that share a similar price point but represent entirely different building types, confirming that a blended model would conflate them.

Annex 5 — Pricing Evidence

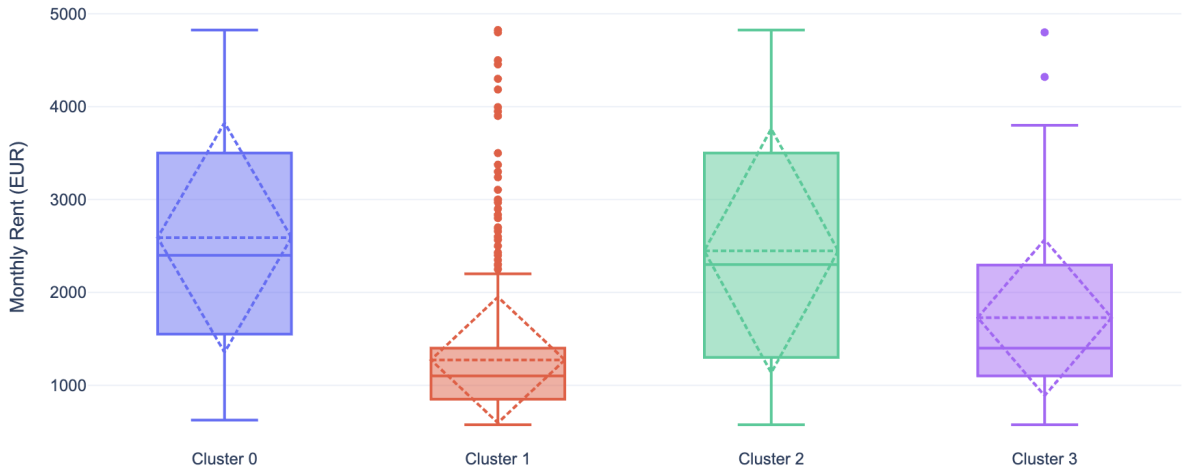


Figure A5: Monthly rent distribution by cluster (EUR). C1 (Compact Traditional) is the tightest and lowest segment — median 1,100, IQR 800–1,250 — confirming it as the volume market. C0 and C2 share a similar median (2,300–2,400) but C2 carries a wider IQR, reflecting greater within-segment price dispersion driven by floor-level variation. C3 (Compact High-Rise) sits between the two price bands with a median near 1,500, occupying a structurally distinct mid-market position. The near-absence of overlap between C1 and the remaining three clusters validates the $K = 4$ segmentation on economic grounds.

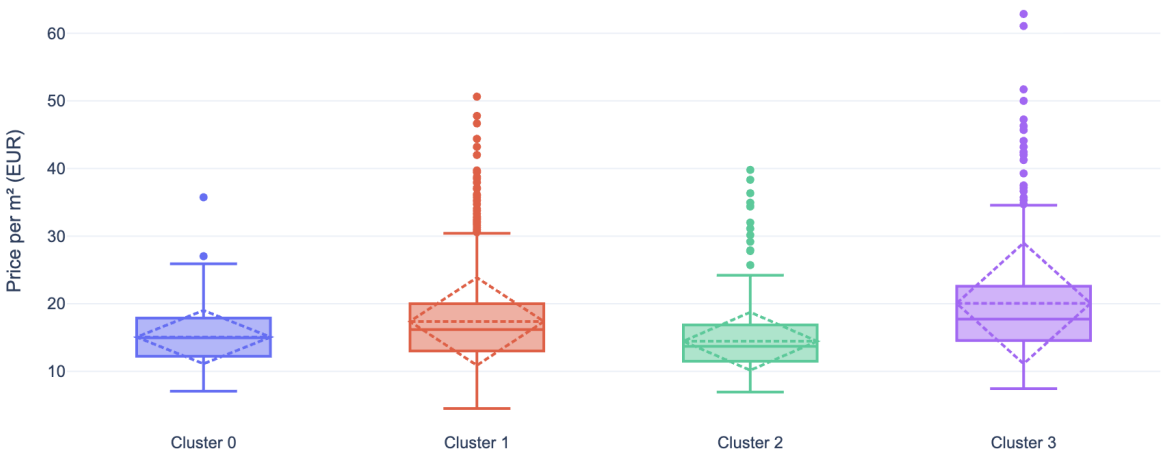


Figure A6: Price per m^2 by cluster (EUR/m^2). Medians converge across all four segments (13–17/m^2), confirming that absolute rent differences are driven by unit size and floor level rather than intrinsic price-per-m^2 premiums. C1 (Compact Traditional) and C3 (Compact High-Rise) show the widest upper tails — consistent with the location-driven heterogeneity captured by the Sq.Mt $\times$ District interaction terms in the regression model. The convergence of medians across clusters provides the economic rationale for modelling interactions rather than level shifts: it is *where* a square metre is located, not how many square metres there are, that drives rent variation within the *Compact Traditional* segment.

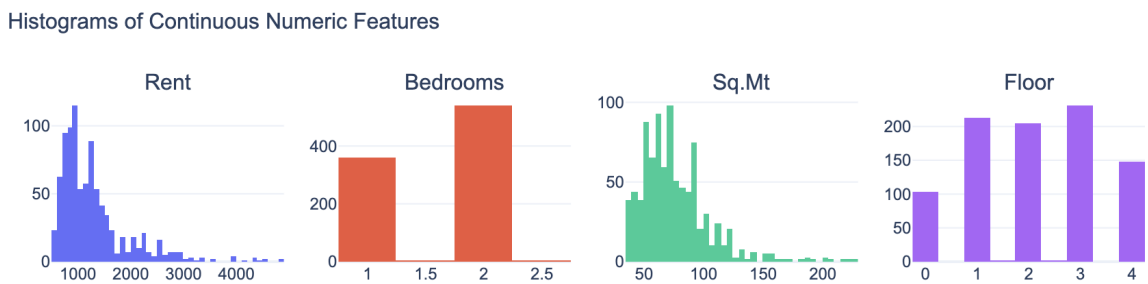


Figure A7: Distribution of continuous numeric features after outlier capping. Rent and Sq.Mt are right-skewed; Bedrooms concentrates at 1–2; Floor peaks at levels 1–3.

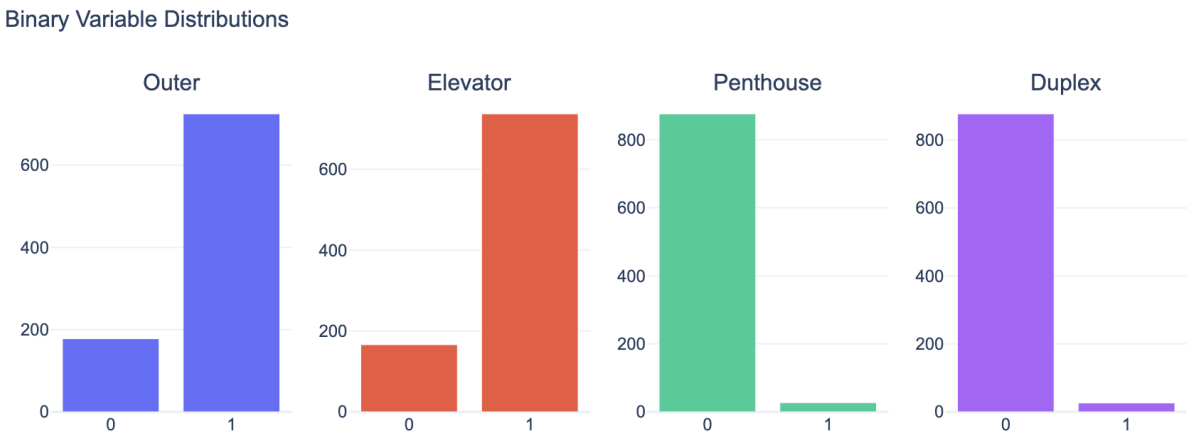


Figure A8: Distribution of binary amenity indicators. Outer (80%) and Elevator (82%) are near-universal in **C1**; Penthouse and Duplex are rare (<5%), justifying mode imputation at 0.

Table A7: District encoding — before and after low-frequency grouping (*districts with <25 listings collapsed into "Other"*).

Before grouping		After grouping	
District	n	**District**	n
Centro	196	Centro	196
Chamberí	90	Chamberí	90
Tetuán	86	Tetuán	86
Salamanca	81	Salamanca	81
Chamartín	75	Chamartín	75
Moncloa	41	Moncloa	41
Ciudad Lineal	39	Ciudad Lineal	39
San Blás	39	San Blás	39
Arganzuela	38	Arganzuela	38
Puente Vallecas	34	Puente Vallecas	34
Hortaleza	30	Hortaleza	30
Fuencarral	29	Fuencarral	29
Carabanchel	27	Carabanchel	27
Retiro	26	Retiro	26
Latina	17	*Other*	72
Usera	17		
Vicálvaro	14		
Villa de Vallecas	14		
Barajas	10		
Total	**903**	**Total**	**903**

Table A8: Initial OLS model — full feature set before variable selection ($n = 406$). Rows highlighted in green are significant at $p < 0.05$; † denotes marginal significance ($p < 0.10$).

Variable	Coef.	Std. Err.	t	p	95% CI lower	95% CI upper
Model fit: $R^2 = 0.628$, *Adj.* $R^2 = 0.607$, $F(21, 384) = 30.80$, $p < 0.001$, $n = 406$, *AIC=6061, BIC=6149*						
const	−2.50	141.93	−0.02	0.986	−281.56	276.56
Bedrooms	−7.95	50.85	−0.16	0.876	−107.93	92.04
Sq.Mt	12.39	0.94	13.22	<0.001 ***	10.55	14.24
Floor	21.08	17.33	1.22	0.225	−12.99	55.14
Outer	113.56	55.93	2.03	0.043 *	3.60	223.52
Elevator	24.24	55.07	0.44	0.660	−84.04	132.52
Penthouse	62.53	125.35	0.50	0.618	−183.94	308.99
Duplex	162.32	162.35	1.00	0.318	−156.89	481.54
District dummies (reference: Carabanchel)						
District_Carabanchel	−130.02	173.68	−0.75	0.455	−471.51	211.46
District_Centro	382.80	118.14	3.24	0.001 **	150.51	615.08
District_Chamartín	239.29	132.48	1.81	0.072 †	−21.20	499.77
District_Chamberí	376.78	128.65	2.93	0.004 **	123.83	629.73
District_Ciudad Lineal	2.37	161.27	0.01	0.988	−314.72	319.46
District_Fuencarral	−79.14	168.02	−0.47	0.638	−409.49	251.21
District_Hortaleza	−25.92	167.67	−0.15	0.877	−355.58	303.75
District_Moncloa	198.83	156.42	1.27	0.204	−108.71	506.37
District_Other	−240.48	135.29	−1.78	0.076 †	−506.47	25.52
District_Puente Vallecas	−160.76	149.96	−1.07	0.284	−455.61	134.08
District_Retiro	264.45	153.95	1.72	0.087 †	−38.24	567.13
District_Salamanca	829.79	127.32	6.52	<0.001 ***	579.46	1080.11
District_San Blás	−222.13	150.03	−1.48	0.140	−517.10	72.85
District_Tetuán	73.75	129.17	0.57	0.568	−180.21	327.71

$***\ p < 0.001$; $**\ p < 0.01$; $*\ p < 0.05$; $\dagger\ p < 0.10$. Skew=1.685; Kurtosis=9.307; DW=2.207.

Table A9: OLS model after dropping clustering variables (Bedrooms, Floor) — rationale: within a single cluster these variables carry minimal variance and add no meaningful predictive value ($n = 406$). Green = $p < 0.05$; † = $p < 0.10$.

Variable	Coef.	Std. Err.	t	p	95% CI lower	95% CI upper
Model fit: $R^2 = 0.626$, *Adj.* $R^2 = 0.608$, $F(19, 386) = 34.01$, $p < 0.001$, $n = 406$, *AIC=6059, BIC=6139*						
const	18.23	129.35	0.14	0.888	−236.09	272.55
Sq.Mt	12.42	0.82	15.15	<0.001 ***	10.81	14.03
Outer	115.99	55.79	2.08	0.038 *	6.30	225.68
Elevator	34.18	54.06	0.63	0.528	−72.11	140.47
Penthouse	91.68	122.96	0.75	0.456	−150.06	333.43
Duplex	170.67	162.10	1.05	0.293	−148.04	489.38
District dummies (reference: Carabanchel)						
District_Carabanchel	−135.75	172.69	−0.79	0.432	−475.28	203.78
District_Centro	385.41	117.98	3.27	0.001 **	153.45	617.38
District_Chamartín	231.53	132.18	1.75	0.081 †	−28.35	491.41
District_Chamberí	374.00	128.53	2.91	0.004 **	121.31	626.70
District_Ciudad Lineal	14.32	160.84	0.09	0.929	−301.90	330.55
District_Fuencarral	−89.65	166.91	−0.54	0.592	−417.80	238.51
District_Hortaleza	−29.87	167.46	−0.18	0.859	−359.12	299.39
District_Moncloa	185.90	155.77	1.19	0.233	−120.37	492.17
District_Other	−239.30	134.78	−1.78	0.077 †	−504.29	25.69
District_Puente Vallecas	−173.88	148.92	−1.17	0.244	−466.67	118.92
District_Retiro	256.59	153.54	1.67	0.095 †	−45.29	558.47
District_Salamanca	828.02	127.21	6.51	<0.001 ***	577.91	1078.14
District_San Blás	−220.92	149.88	−1.47	0.141	−515.60	73.77
District_Tetuán	76.15	128.78	0.59	0.555	−177.05	329.34

$***\ p < 0.001$; $**\ p < 0.01$; $*\ p < 0.05$; $\dagger\ p < 0.10$. Skew=1.649; Kurtosis=9.019; DW=2.205.

Table A10: Variance Inflation Factors after dropping clustering variables. VIF ≥ 5 flagged in amber; no variable exceeds the hard threshold of 10.

Variable	VIF
Physical features	
Sq.Mt	8.68
Outer	5.90
Elevator	5.23
Penthouse	1.07
Duplex	1.08
District dummies	
District_Centro	2.75
District_Salamanca	2.18
District_Chamartín	1.80
District_Chamberí	1.70
District_Tetuán	1.65
District_Other	1.61
District_San Blás	1.45
District_Retiro	1.42
District_Moncloa	1.32
District_Puente Vallecas	1.29
District_Ciudad Lineal	1.29
District_Fuencarral	1.28
District_Hortaleza	1.25
District_Carabanchel	1.17

Threshold: VIF < 5 acceptable; VIF ≥ 10 requires removal.

Table A11: Final OLS model — Sq.Mt $\times$ District interactions ($n = 406$, $df_{\text{model}} = 10$). All retained predictors significant at $p < 0.05$. Condition number = 429, well below the collinearity threshold of 1,000.

Variable	Coef.	Std. Err.	t	p	95% CI lower	95% CI upper
Model fit: $R^2 = 0.644$, *Adj.* $R^2 = 0.635$, $F(10, 395) = 71.53$, $p < 0.001$, $n = 406$, *AIC=6020, BIC=6065*						
Base predictors						
const	375.39	63.26	5.94	<0.001 ***	251.04	499.75
Sq.Mt	7.38	0.95	7.78	<0.001 ***	5.52	9.25
Outer	107.27	51.84	2.07	0.039 *	5.37	209.18
District_Other	−214.98	79.51	−2.70	0.007 **	−371.30	−58.65
District_Puente Vallecas	−199.91	99.25	−2.01	0.045 *	−395.03	−4.80
Sq.Mt $\times$ District interactions (additional per m^2 vs. baseline)						
SqMt $\times$ Centro	+6.03	0.73	8.31	<0.001 ***	4.60	7.46
SqMt $\times$ Chamartín	+3.79	0.99	3.85	<0.001 ***	1.85	5.73
SqMt $\times$ Chamberí	+5.78	0.96	6.02	<0.001 ***	3.90	7.68
SqMt $\times$ Moncloa	+3.22	1.41	2.28	0.023 *	0.44	5.99
SqMt $\times$ Retiro	+4.26	1.20	3.57	<0.001 ***	1.92	6.61
SqMt $\times$ Salamanca	+10.85	0.83	13.12	<0.001 ***	9.22	12.47

$***\, p < 0.001$; $**\, p < 0.01$; $*\, p < 0.05$. Skew=1.738; Kurtosis=9.250; DW=2.192; Cond. No.=429.

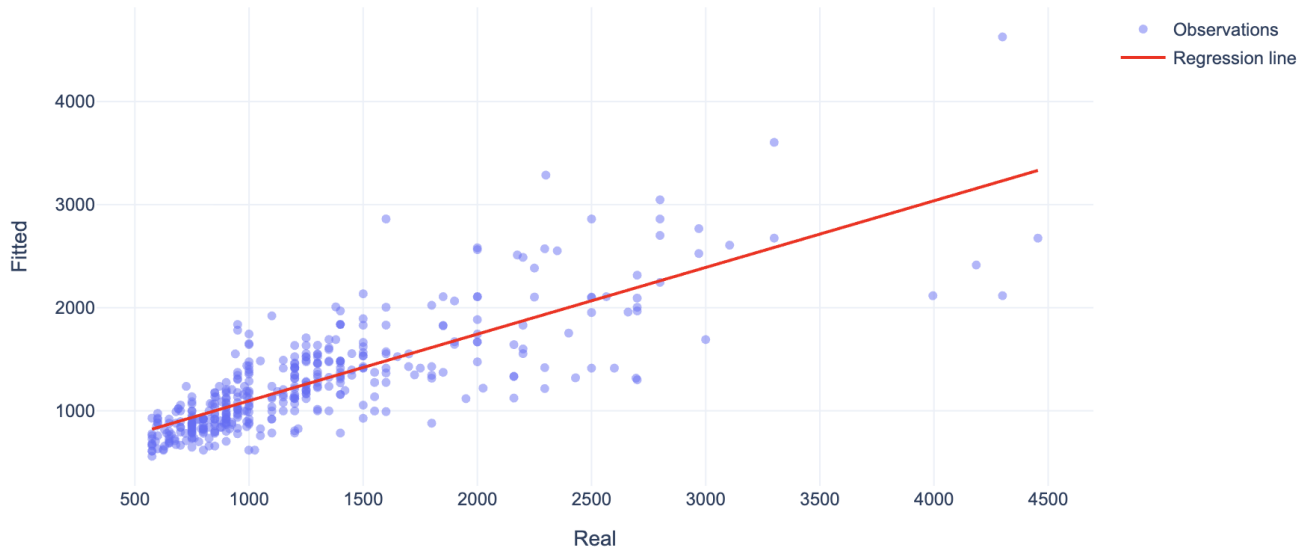


Figure A9: Real vs. fitted values — training set ($n = 406$). Points cluster tightly along the diagonal... Scatter above €2,000 reflects high-specification properties whose prestige-street effects lie outside the model's observable features.

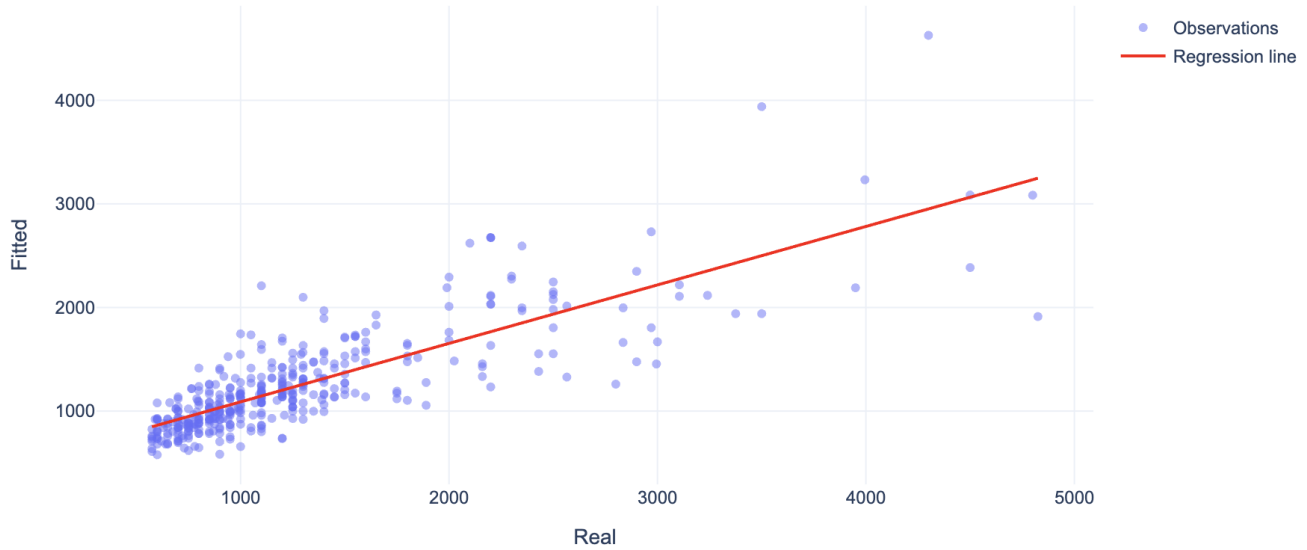


Figure A10: Real vs. fitted values — test set ($n = 203$). The regression line maintains a consistent slope with no systematic tilt, confirming the model generalises to unseen data. Scatter increases above 2,000, where prestige-street effects and interior specification push actual rents beyond what district and square meters alone can explain.

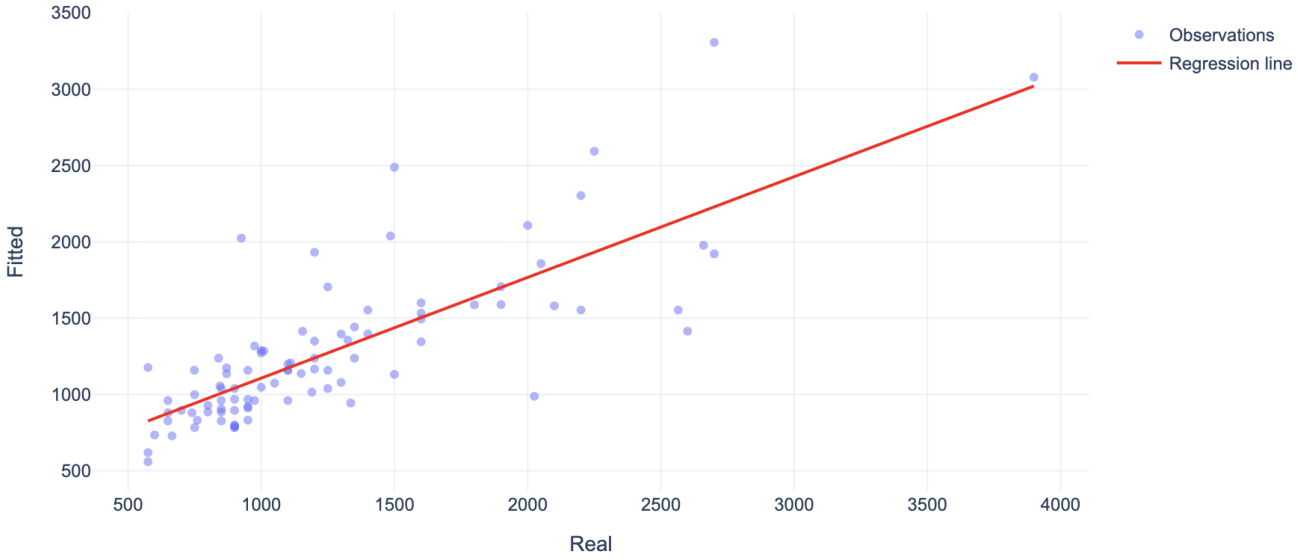


Figure A11: Real vs. fitted values — reserved hold-out ($n = 41$). Despite never being seen at any stage of training or feature selection, the regression slope remains consistent with the training and test splits. The two high-leverage points above 2,500 confirm the upper-end under-prediction pattern is structural, not a product of the training sample.

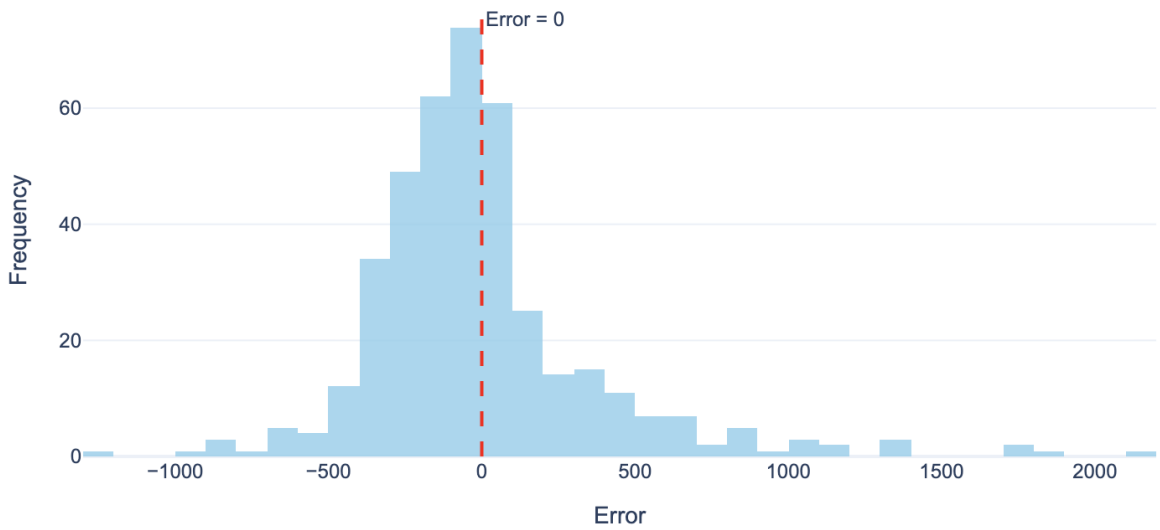


Figure A12: Residual distribution — training set ($n = 162$). The distribution centers tightly on zero with no systematic shift, confirming the model is unbiased across the majority of the segment. The right tail extending to +2,100 and the near-absence of a symmetric left tail indicate the model under-predicts selectively at the top end — consistent with the Skew = 1.738 and Kurtosis = 9.250 reported in the OLS diagnostics.

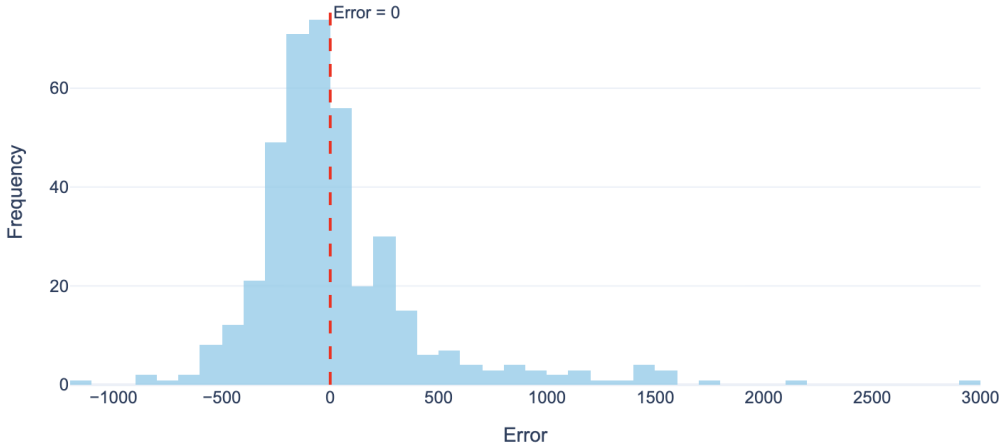


Figure A13: Residual distribution — test set ($n = 203$). The distribution mirrors the training pattern: mass concentrated within $\pm$500 of zero with no leftward shift, indicating the model does not systematically over-predict on unseen data. The isolated bars beyond $+$1,500 reinforce that large errors are confined to a handful of outlier properties rather than distributed across the segment, which explains the RMSE–MAE gap without inflating the typical error.

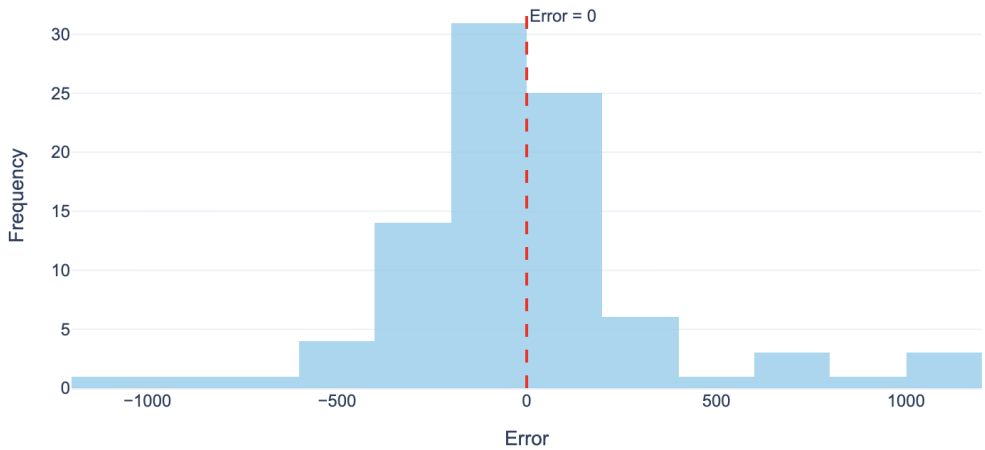


Figure A14: Residual distribution — reserved hold-out. The bulk of errors concentrates within $\pm$300 of zero with no systematic bias, confirming the model does not persistently over- or under-predict. The right tail extending to $+$2,100 reflects the same structural ceiling identified in the real vs. fitted plots: a small number of high-specification properties whose rents exceed what observable features can explain.